



What is ZigBee?

ZigBee is a set of standards for low-power, low data-rate networks that operate over small distances. ZigBee networks work even in congested radio environments. Up to 255 devices can be connected in a network, a maximum of 30 m away from each other.

How does ZigBee work?

ZigBee is a wireless network of devices that communicate with each other. Modest amounts of data are shot over small distances. ZigBee devices can be arranged in a *Star* network, where devices report to a ZigBee Controller; a *Mesh* network, where all devices are at the same level and 'co-operate' with each other; or a *Clustered Tree* network, which combines the best of both.

Why ZigBee?

ZigBee is targeted towards Wireless Personal Area Networks (WPANs), so a ZigBee network is cheaper and simpler to implement than a Bluetooth or Wi-Fi network. A typical ZigBee node will require two to 10 per cent of the software a Bluetooth node would require. Extending a ZigBee network is easy—you don't need to re-wire any of the devices to add another ZigBee node.

Where would I see a ZigBee network?

ZigBee can be used in homes and offices to remote control every device possible. In factories, they can be used to automate manufacturing tasks and asset management besides monitoring heat and energy usage to optimise energy consumption.

Who thought it up?

The ZigBee Alliance, a consortium of companies, published the ZigBee standard. They monitor and control products that seek to employ the standard.

When was ZigBee started?

Although in development since 1999, the ZigBee standard was ratified by the ZigBee Alliance in December 2004. In June 2005, the ZigBee specification was thrown open to the public.

Shir's paper only discusses his mathematical model; there has been no effective implementation as yet. Shir, however, plans to release a simple example program soon.

"Our focus is to immunise the whole network, not to clean individual computers or fix what is already broken," Shir said on the Web edition of *Nature*. In a paper titled "Distributive immunisation of networks against viruses using the 'honey-pot' architecture," Shir and colleagues have made a proposal "to propagate the immunisation agent as an epidemic."

The current problem is that if a cure is sent over the same network as that on which the virus is spreading, there is bound to be a lag. The virus will affect nodes (computers) before the cure gets to them. Shir and colleagues therefore propose the development of dedicated "honeypot"—or attractor—computers, distributed all over the Net. The honeypots would attract a virus (to which the honeypot would appear as a regular computer), analyse it automatically, and distribute a countermeasure.

The honeypots would be linked to each other via a dedicated, secure network. This way, once one honeypot has attracted and analysed a virus, all the others would immediately know about the infection—the existence of the "honeypot network" would enable the countermeasure system to be faster than the virus. That's where the real trick lies. Each honeypot would act as a hub sending out healing countermeasure code. This countermeasure would spread across all of the network—in theory, possibly even all of the Internet: simulations have shown that the larger the network is, the more efficient this scheme would be.

WHAT'S THAT SONG?

Google Eases Music Search

Google is tailoring its search engine to better address music search. A search for an artist will now bring up song titles, album art, links to lyrics and reviews, and advertisements for places where the music can be purchased. Items that can be purchased will have links to merchants for online ordering or downloading. The online sources include Apple, RealNetworks and eMusic, and retailers selling CDs including Amazon, Wal-Mart and others. For the CD retailers, Google receives data feeds of the inventory and only returns a link to a store if the item is available.

Search results will include links to supplemental Google Web pages with more information about the music, including names of tracks on a CD and other CDs a band or artist has released. Google also will provide snippets of reviews from sites on the Web, as well as links to those sites.

The company has, of course, constantly been improving its search service to match user needs. Marissa Mayer, the company's vice president for search products and user experience, admitted "one of the things Google wasn't good at" and that consumers wanted was music.

There's more to the move than just better search results: Google's

current focus on music comes at a time when more companies are selling and distributing music online. Mike McGuire, research director for Gartner Inc., said Google would be a natural place for consumers to look when they're trying to think of songs or shop for music. "It gets people to move from CDs to online."

Google's improvement of its music search capabilities is part of the always-on war with its rivals.



Yahoo! launched its Yahoo! Music service earlier this year, and tied it with its blogs and other communication tools. And with the proliferation of online music content, more companies are enhancing their search capabilities: Apple's iTunes, for example, allows users to search its database by artist or title, or within various genres. Online libraries such as Rhapsody and Napster also offer searches of their databases. And services such as 411-Song sell song identification: you can hold a cell phone up to a speaker playing a tune, and you get the name of the song you played! ☐

Compiled by Jyotsna Rege, Nimish Chandiramani, Ram Mohan Rao and Renuka Rane